

INTERFACE CHALLENGE

Problem Statement

An insurance company wants to model different categories of insurance using an interface.

All insurance policies share the following behaviors:

Get policy holder name

Calculate premium

Display policy details

Use the following interface:

```
public interface InsurancePolicy
{
    String getPolicyHolder();
    double calculatePremium();
    void displayPolicyDetails();
}
```

Implement the following three types of insurance:

Health Insurance

Premium = Base Amount + (Age × 100)

Attributes:

policyHolder (String)

baseAmount (double)

age (int)

Vehicle Insurance

Premium = Vehicle Value × 0.05

Attributes:

policyHolder (String)

vehicleValue (double)

Life Insurance

Premium = (Coverage Amount / Term Years) × 0.02

Attributes:

policyHolder (String)

coverageAmount (double)

termYears (int)

Input Format

First line: choice (Integer)

1 → Health Insurance

2 → Vehicle Insurance

3 → Life Insurance

Depending on the choice:

If choice = 1 (Health Insurance)

policyHolder (String)

baseAmount (double)

age (int)

If choice = 2 (Vehicle Insurance)

policyHolder (String)

vehicleValue (double)

If choice = 3 (Life Insurance)

policyHolder (String)

coverageAmount (double)

termYears (int)

Constraints:

$1 \leq \text{choice} \leq 3$

$1 \leq \text{age} \leq 100$

$1 \leq \text{termYears} \leq 50$

$1.0 \leq \text{baseAmount} \leq 10^7$

$1.0 \leq \text{vehicleValue} \leq 10^8$

$1.0 \leq \text{coverageAmount} \leq 10^9$

policyHolder length ≤ 100 characters

All monetary values are positive

Output Format

Policy Type: <Insurance Type>

Policy Holder: <policyHolder>

Premium: <calculatedPremium formatted to 2 decimal places>

Sample Test Cases (minimum 2)

Test Case	Sample Input	Sample Output
1	1 Charlie 5000	Policy Type: Health Insurance Policy Holder: Charlie Premium: 8000.00

Test Case	Sample Input	Sample Output
	30	
2	2 Arjun 400000	Policy Type: Vehicle Insurance Policy Holder: Arjun Premium: 20000.00
3	3 Meera 1000000 20	Policy Type: Life Insurance Policy Holder: Meera Premium: 1000.00

METHOD OVERLOADING CHALLENGE

Problem Statement

The GPA (Grade Point Average) can be calculated in four different ways:

1. From marks in one subject

$$\text{GPA} = \text{marks} / 10.0$$

2. From marks in three subjects

$$\text{GPA} = (\text{marks1} + \text{marks2} + \text{marks3}) / 30.0$$

3. From percentage

$$\text{GPA} = \text{percentage} / 20.0$$

4. From marks and extra credit

$$\text{GPA} = (\text{marks} + \text{extraCredit}) / 10.0$$

Write a Java program using method overloading to calculate GPA based on user choice.

Input Format

First line: Integer choiceo

1 → GPA from one subject

2 → GPA from three subjects

3 → GPA from percentage

4 → GPA from marks and extra credit

Depending on the choice:

If choice = 1

marks (int)

If choice = 2

marks1(int)

marks2(int)

marks3 (int)

If choice = 3

percentage (double)

If choice = 4

marks (int)

extraCredit (int)

Constraints: $1 \leq \text{choice} \leq 4$ $0 \leq \text{marks} \leq 100$ $0 \leq \text{percentage} \leq 100$ $0 \leq \text{extraCredit} \leq 20$ **Output Format**

The calculated GPA formatted to two decimal places.

Sample Test Cases (covering all inputs)

Test Case	Sample Input	Sample Output
1	1 85	8.50
2	2 90 80 70	8.00
3	3 88.0	4.40
4	4 75 5	8.00